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ISBA 2403: Machine Learning with Python

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Wine Quality Written Report

Background Section

Have you ever thought of how the flavor and quality of the wine might shape your overall experience? The goal of this project is to analyze and predict wine quality using machine learning techniques within the wine production industry. Evaluating wine quality is an important task because it directly affects product pricing, consumer satisfaction, and brand reputation. Traditionally, wine quality has been assessed by expert tasters who evaluate sensory characteristics such as flavor, color, and aroma. However, this approach can be subjective, time-consuming, and costly, which has led researchers and industry professionals to explore more objective methods for evaluating wine quality using data.

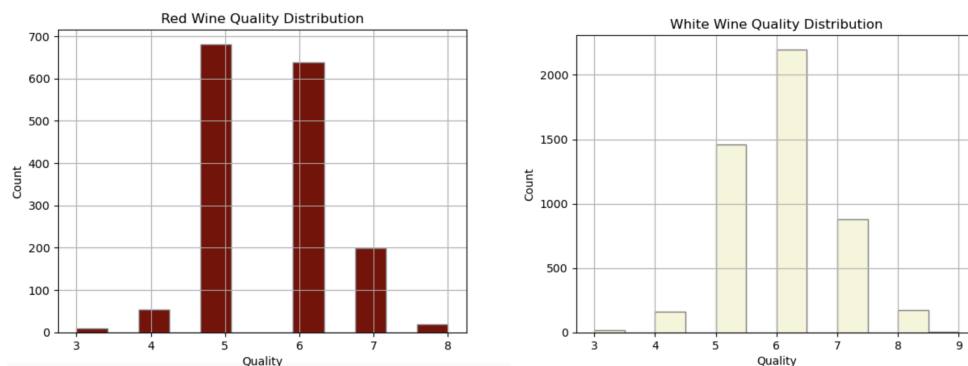
Machine learning provides a data-driven solution by identifying patterns in measurable chemical properties of wine, including acidity, pH level, alcohol content, and sugar concentration. In this project, the machine learning task is regression, where the model predicts a numerical wine quality score based on these chemical attributes. By learning relationships between chemical measurements and expert quality ratings, the model can estimate the quality of new wine samples. This approach can reduce subjectivity while providing faster and more consistent evaluations, helping wine producers monitor quality during production, identify key factors that influence higher-quality wines, and make better decisions throughout the winemaking process.

Dataset Overview

The dataset used in this project is the **Wine Quality Dataset** from the UCI Machine Learning Repository, donated in 2009, including red and white Vinho Verde wine samples from the north of Portugal. It contains laboratory measurements of different chemical properties of wine along with quality ratings given by professional wine tasters. These ratings are based on sensory factors such as taste, aroma, and overall balance. Since the dataset combines objective chemical data with subjective quality scores, it is well-suited for applying machine learning to predict wine quality.

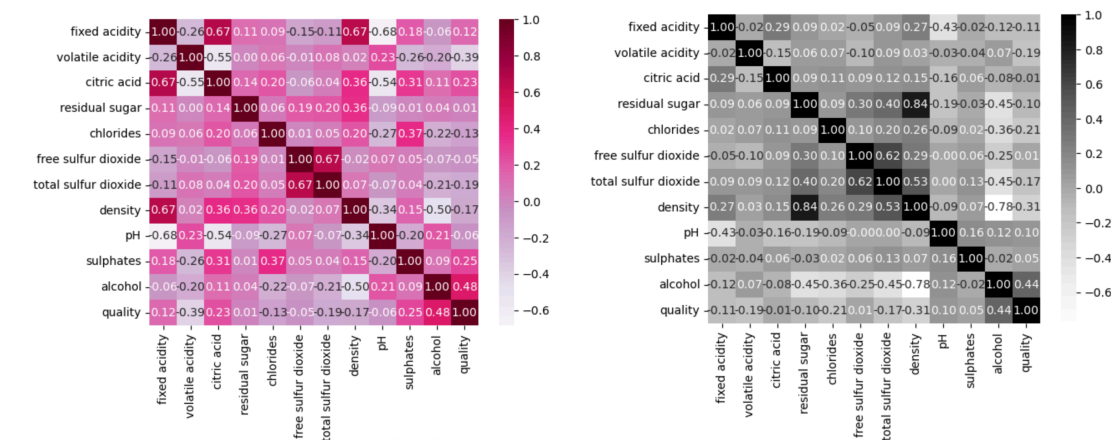
The dataset includes two parts: one for red wine and one for white wine. The red wine dataset has 1,599 samples, while the white wine dataset has 4,898 samples, providing a total of 6,497 wine observations. Each sample represents a different wine tested in a laboratory. Both datasets contain the same features, which allows them to be analyzed separately or combined into one larger dataset. The wine sample includes 11 features such as acidity levels, sugar content, pH, density, and alcohol content. These variables describe the chemical composition of the wine and influence its taste, stability, and overall quality. The dataset also includes a target variable, quality, which is a score from 0 to 10 given by wine experts, and in this dataset, most wines have quality scores between 3 and 8.

Figures 1 and 2



Before building the model, the data must be explored and prepared. We created bar charts, heatmaps, and scatterplots to analyze the data we are working with. Figures 1 and 2 show the wine quality distribution of both the red and white wines. In both of these datasets, most of the quality scores fall between about 5 and 7, with relatively few wines rated extremely low or extremely high. These distributions tell us that the dataset is fairly concentrated around mid-range quality scores, which is useful to know before building our predictive models.

Figures 3 and 4



In Figures 3 and 4, we examined the correlation matrices to understand relationships between the chemical features and wine quality. Looking specifically at the quality row, we can see that alcohol has one of the strongest positive correlations with wine quality in both datasets. For red wine, sulphates also show a noticeable positive relationship with quality, while in white wine, we see stronger relationships involving sulfur-related variables and density. These correlations helped guide our later modeling and feature interpretation.

Model Section

The goal of this project is to predict wine quality using physicochemical attributes from the wine dataset. Because the target variable quality is numeric, this task is formulated as a

supervised regression problem. To ensure a robust modeling approach, we implemented three different ensemble machine learning models: Gradient Boosting Regressor, Random Forest Regressor, and Bagging Regressor. These models were chosen because ensemble methods often provide strong predictive performance by combining multiple decision trees to reduce variance and improve generalization.

Before training the models, the dataset was split into training and testing sets using an 80/20 split, with a fixed random seed (`random_state = 42`) to ensure reproducibility. Feature variables were standardized using `StandardScaler`, which rescales the features to have a mean of 0 and a standard deviation of 1. Although tree-based models do not strictly require scaling, applying standardization helps maintain consistency across models. To improve model performance and avoid overfitting, we used `GridSearchCV` with 10-fold cross-validation to tune model hyperparameters. Grid search systematically evaluates multiple hyperparameter combinations and selects the set that produces the best validation performance. The models were evaluated using negative mean squared error (`neg_mean_squared_error`) during cross-validation. This metric measures the squared difference between predicted and actual values, where, when changed to a positive, lower values indicate better predictive performance.

The first model we used was a Gradient Boosting Regressor. We tested different combinations of hyperparameters, including `n_estimators` (100, 200, 300), which controls the number of trees, `max_depth` (2, 3), which limits the depth of each tree, `learning_rate` (0.01, 0.05), which controls how much each tree contributes to the model, and `subsample` (0.7, 0.8, 0.9), which adds randomness.. After running the grid search on the scaled training data (`Xr_train_scaled` and `yr_train`), the best combination of parameters was selected and used as the

final model to predict wine quality. The optimal hyperparameters for both datasets can be seen below:

Dataset	n_estimators	max_depth	learning_rate	subsample
Red Wine	300	3	0.05	0.8
White Wine	300	3	0.05	0.7

We also used a Random Forest Regressor to predict wine quality. The model was initialized with `bootstrap = True` and `random_state = 42` for reproducibility. We tested different hyperparameters, including `n_estimators` (200, 400) for the number of trees, `max_depth` (3, 5, 8) to control how deep the trees can grow, `max_features` ('sqrt', 'log2') to decide how many features are considered at each split, `min_samples_split` (5, 10) and `min_samples_leaf` (2, 4), to ensure that splits only occur when there are enough observations.. The optimal hyperparameters can be seen below:

Dataset	n_estimators	max_depth	max_features	min_samples_split	min_samples_leaf
Red Wine	400	8	sqrt	5	2
White Wine	200	8	sqrt	5	2

Lastly, we used a Bagging Regressor with a Decision Tree Regressor as the base model to predict wine quality. The model was initialized with `random_state = 42` for reproducibility, as well as `min_samples_split = 10` and `min_samples_leaf = 4` to control overfitting and to keep base learners simple. We tested different hyperparameters, including `n_estimators` (100, 300), which controls the number of trees in the ensemble, `max_samples` (0.6, 0.8), which determines the

proportion of the training data used to train each tree, and `max_features` (0.7, 1.0), which controls the fraction of features used for each tree. The best parameter combination was selected and used as the final Bagging model for predicting wine quality.

Dataset	n_estimators	max_samples	max_features
Red Wine	300	0.6	1.0
White Wine	300	0.6	1.0

Overall Results

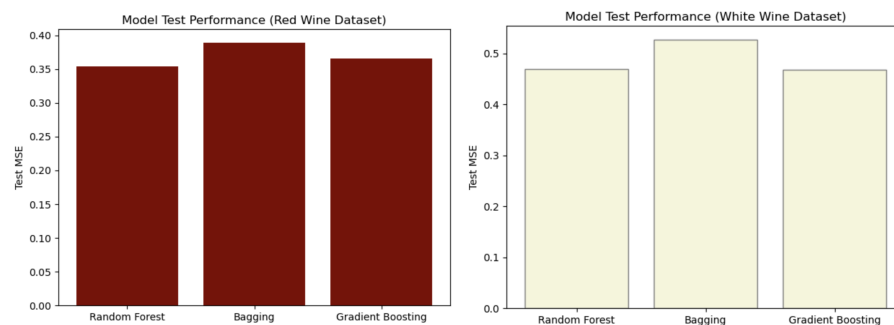
After running the code, we found a few interesting results. There were no missing values detected amongst all of the observations in both datasets. As such, we did not have to clean the data before we began the modeling process. Out of the features we recorded, we chose quality as our target, and the rest of the 11 features became our predictors. However, we did notice that the datasets were relatively small compared to those of other ML datasets; as such, we were aware that these small datasets may influence model generalization.

Cross Validation Results

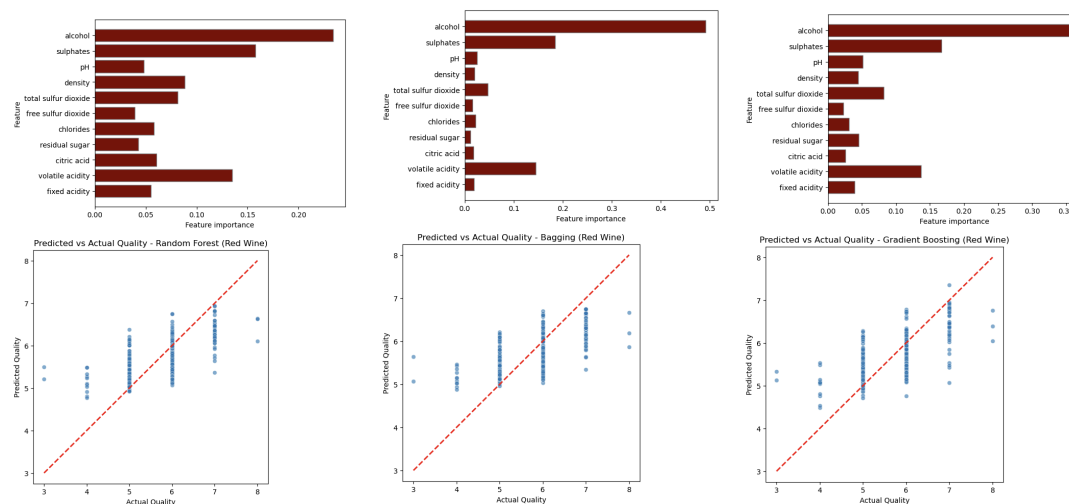
As specified above, we used three models for each of the wine datasets. When looking at the red wine models, the random forest regressor achieved the lowest cross-validation error among the three models we tested. The optimal hyperparameters produced a training set cross-validation MSE of 0.380. The bagging regressor performed similarly, with a cross-validation MSE of 0.408. The gradient boosting regressor achieved a cross-validation MSE of 0.387. As we can see, all three models produced relatively similar results with different hyperparameters. However, random forest achieved the lowest error, which indicates that this model had a slightly stronger predictive performance for the red wine dataset.

For the white wine dataset, we found that random forest again produced the best cross-validation performance. The optimal hyperparameters for random forest achieved a training set cross-validation MSE of 0.469, the lowest among the models. The bagging regressor outputted a cross-validation MSE of 0.531. The gradient boosting regressor produced a cross-validation MSE of 0.471. Overall, random forest consistently showed the lowest prediction error across both datasets, which suggests that it may be the most reliable model for predicting wine quality based on the available chemical features. However, it is important to note that the differences in cross-validation MSEs across the models were relatively small. This means that all models captured similar predictive patterns within the dataset, but the random forest model achieved a slightly better performance.

Figures 5 and 6:



Figures 7, 8, 9:



Figures 10, 11, 12:

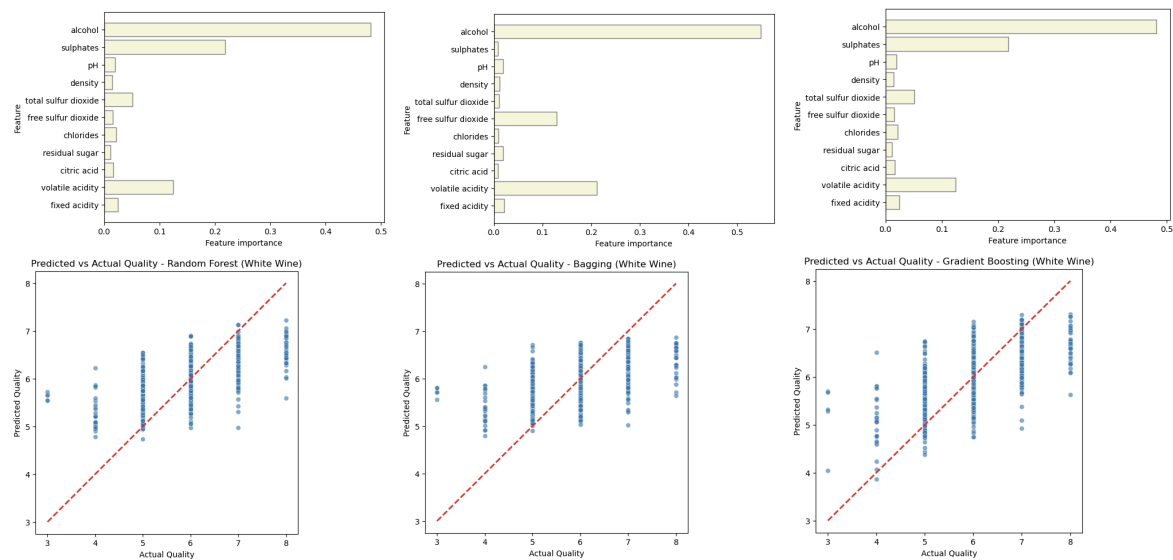


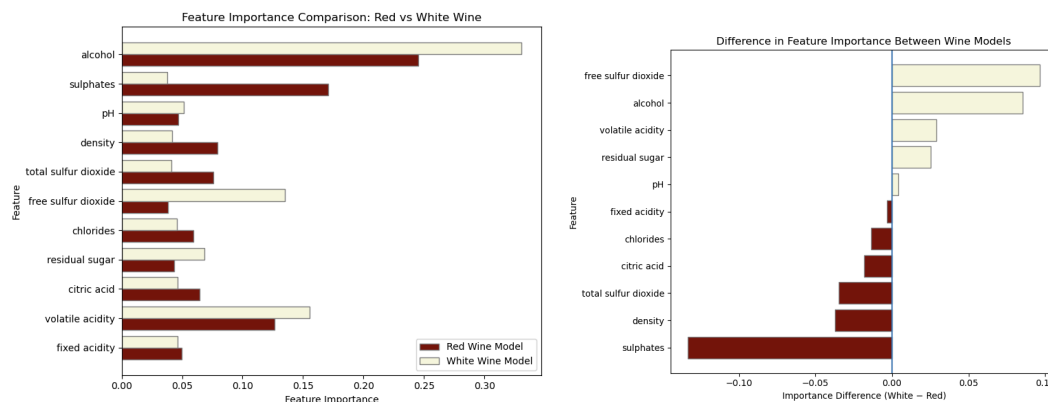
Figure 5 and 6 shows the test set mean squared error (MSE) for the three ensemble models applied to both the red and white wine datasets, respectively. For the red wine dataset, the random forest model achieved the lowest prediction error, followed by the gradient boosting regressor and bagging regressor. We see a similar pattern in the white wine dataset, but gradient boosting produces the lowest test MSE by 0.001 above the test MSE for random forest. The difference is so small that we decided to choose random forest as the best model, because it was the most consistent across training and testing sets. Since lower MSE values indicate better predictive performance, these results suggest that random forest provides the most accurate predictions for wine quality across both datasets.

Figures 7, 8, and 9 show the feature importance rankings and predicted versus actual quality comparisons for the red wine dataset across the three ensemble models. The feature importance plots show which chemical attributes contribute most strongly to the prediction of wine quality. Across all models, alcohol consistently appears as the most influential predictor, followed by variables such as sulphates and volatile acidity. The predicted versus actual scatter

plots illustrate how closely each model's predictions align with the true quality values. Points closer to the diagonal reference line represent more accurate predictions, and the random forest model shows slightly tighter clustering around this line, which is consistent with its lower prediction error.

Figures 10, 11, and 12 present the feature importance rankings and predicted versus actual quality comparisons for the white wine dataset. Similar to the red wine results, alcohol continues to remain as one of the most influential predictors across the models. Other chemical properties, such as sulfur dioxide levels and volatile acidity, also contribute to predicting wine quality. Predicted versus actual plots demonstrate that the models capture the general relationship between predicted and actual wine quality values, although some variation from the diagonal reference line remains. This indicates that while the models successfully identify overall trends within the dataset, predicting exact wine quality scores continues to be challenging.

Figures 13 and 14:



Figures 13 and 14 compare the feature importances for red wine and white wine. Both wines show alcohol as the most important feature, and volatile acidity as another important one. Looking at Figure 14, we can see that sulphates affect red wines significantly more than they affect white wines, and free sulfur dioxide affects white wines more than it affects red wines.

Conclusion/Recommendations

To summarize, this project applied machine learning models, such as Gradient Boosting, Random Forest, and Bagging, to predict wine quality using chemical attributes from the UCI Wine Quality dataset. After running the data, **the Random Forest model** produced the most accurate predictions for both red and white wine datasets. The findings highlight that certain features, particularly alcohol content and acidity-related variables, play an important role in determining wine quality.

Based on our analysis, we recommend that wine producers carefully manage sulphates in red wine, since they have a strong effect on quality, while also improving fermentation control to maintain good alcohol levels and acidity balance. For both red and white wines, higher alcohol content and balanced acidity were linked to better quality, showing that grape maturity and controlled production are important. Our findings also suggest that wine sellers can improve marketing by using descriptions such as “full-bodied,” “structured,” and “bright acidity” to highlight qualities associated with higher-rated wines.

Looking at the overall project, it demonstrates how machine learning can provide a more efficient way to evaluate wine quality, allowing producers to better monitor the production process and make improvements when needed.